Sri Lanka Institute of Information Technology



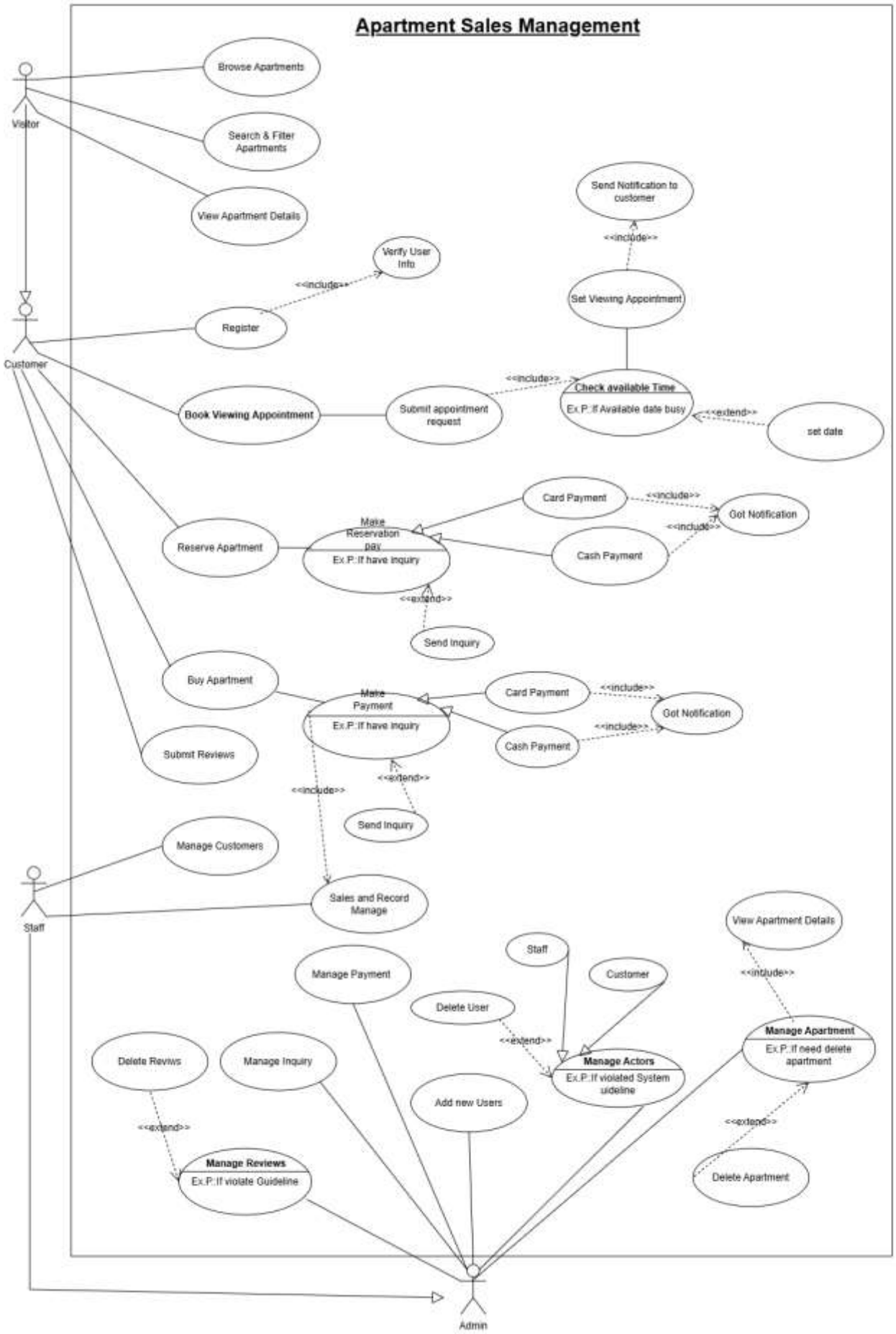
SE2030 - SoftwareEngineering
Year 1, Semester 1- 2025

Lab Submission 3

Group 26-MTR-IT2011-12

IT Number	Name
IT25100393	Pathirannehelage T. J. T.
IT25102539	Ransith K. T.
IT25103131	Dharmapala H.K.S.
IT25103860	Kumarge P. T. A. M.
IT25103871	Jathusayinie K.
IT25103887	Kahanda I. M.

Use Case Diagram



Use Case Scenarios

- IT25100393 Pathirannehelage T. J. T.

Number	UC-01	
Name	Payment and Transaction Management	
Summary	Customers make reservation or purchase payments while finance staff verifies payments and manages payment records.	
Priority	5	
Preconditions	Customer is logged in and has an active reservation or apartment purchase record.	
Postconditions	Payment transaction is recorded, payment status is updated, and the outstanding balance is recalculated.	
Primary Actor	Customer	
Secondary Actor	Financial Manager , System	
Trigger	Customer selects Make Payment for a reservation or apartment purchase.	
Main Scenario	**Step**	**Action**
	1	Customer opens the payment section.
	2	The system displays the reservation or purchase details and amount due.
	3	Customer selects a payment method.
	4	Customer enters the payment amount.
	5	Customer uploads payment evidence such as a bank transfer slip if required.
	6	Customer submits the payment details.
	7	The system records the transaction with Pending status.
	8	Finance staff views the pending payment.
	9	Finance staff verifies the payment details and evidence.
	10	Finance staff confirms the payment.
	11	The system updates the payment status.
	12	The system recalculates the outstanding balance.
	13	The system generates a payment receipt and updates payment history.
Extensions	**Step**	**Action**
	5a	If payment evidence is missing, the system requests the customer to upload it.
	9a	If payment evidence is invalid, finance staff rejects the payment.
	10a	If the payment amount is less than the total amount due, the system records it as Partially Paid.
	11a	If an installment plan is used, the payment is assigned to the relevant installment.
	12a	If the payment due date has passed, the system may mark the payment as Overdue.
Open Issues:	Should online card payments be verified automatically or still require finance staff approval?	

- IT25102539 Ransith K. T.

Number	UC-02	
Name	Inquiry, Review, Notification and Favorites Management	
Summary	Customers submit inquiries, save favorite apartments, submit reviews and receive notifications while administrators manage inquiries and moderate reviews.	
Priority	4	
Preconditions	Customer is logged into the system.	
Postconditions	Customer engagement information is stored and the relevant status is updated.	
Primary Actor	Customer	
Secondary Actor	Administrator, System	
Trigger	Customer selects an inquiry, favorite, review or notification-related function.	
Main Scenario	**Step**	**Action**
	1	Customer opens the customer engagement section.
	2	Customer selects Submit Inquiry.
	3	The system displays the inquiry form.
	4	Customer selects an inquiry category and priority level.
	5	Customer enters the inquiry details.
	6	Customer submits the inquiry.
	7	The system records the inquiry with Pending status.
	8	Administrator views the inquiry from the Administrator Dashboard.
	9	Administrator reviews and replies to the inquiry.
	10	Administrator changes the inquiry status to In Progress.
	11	After resolving the issue, Administrator changes the status to Resolved.
	12	The system sends a notification to the customer.
	13	Customer may add apartments to the Favorites list.
	14	Customer may submit apartment reviews and ratings.
	15	Administrator reviews submitted reviews.
	16	Administrator approves the review and the system publishes it.
	17	Customer receives notifications related to inquiries, appointments, reservations and payments.
Extensions	**Step**	**Action**
	4a	If no suitable inquiry category is available, the customer selects Other.
	9a	Administrator may request additional information from the customer.
	11a	If the issue cannot yet be resolved, the inquiry remains In Progress.
	13a	Customer may remove an apartment from the Favorites list.
	15a	If a review contains inappropriate content, Administrator rejects it.
	17a	If there are no new notifications, the system displays no new notifications.
Open Issues:	Should customers be allowed to edit reviews after administrator approval?	

- IT25103131 Dharmapala H.K.S.

Number	UC-03	
Name	Apartment Sales and Record Management	
Summary	Staff manages customer leads and converts confirmed reservations into completed apartment sales while maintaining sales records and reports.	
Priority	5	
Preconditions	Customer has a confirmed reservation and the required payment has been verified.	
Postconditions	Sale is recorded, apartment status is updated to Sold, and the sales history is updated.	
Primary Actor	Staff	
Secondary Actor	Administrator, System	
Trigger	Staff selects Convert Reservation to Sale for a confirmed reservation.	
Main Scenario	**Step**	**Action**
	1	Staff opens the Apartment Sales Management section.
	2	The system displays customer leads and confirmed reservations.
	3	Staff selects the relevant customer or reservation.
	4	The system displays customer, apartment, reservation and payment information.
	5	Staff verifies that the reservation is confirmed.
	6	Staff verifies that the required payment has been approved.
	7	Staff selects Convert Reservation to Sale.
	8	The system requests confirmation.
	9	Staff confirms the sale.
	10	The system creates the apartment sales record.
	11	The system links the customer, apartment, reservation and payment records.
	12	The system changes the apartment status to Sold.
	13	The system updates the customer purchase history.
	14	Staff can search and view the completed sales record.
	15	Administrator can access sales reports.
Extensions	**Step**	**Action**
	5a	If the reservation is not confirmed, the system prevents the sale.
	6a	If the required payment has not been verified, the system prevents the sale.
	8a	If staff cancels the confirmation, no sale record is created.
	10a	If a sales record already exists for the reservation, the system prevents duplicate creation.
	15a	If no sales records match the selected report criteria, the system displays no matching records.
Open Issues:	Should completed sales records be editable by staff or only by administrators?	

- IT25103860 Kumarge P. T. A. M.

Number	UC-04	
Name	Apartment Listing Management	
Summary	Staff manages apartment projects and units while customers and visitors search and view apartment information.	
Priority	5	
Preconditions	Staff member is logged in and has permission to manage apartment listings.	
Postconditions	Apartment information is successfully added or updated and the latest availability is shown to users.	
Primary Actor	Staff	
Secondary Actor	Customer, Visitor, System	
Trigger	Staff selects Apartment Listing Management from the dashboard.	
Main Scenario	**Step**	**Action**
	1	Staff opens the Apartment Listing Management section.
	2	The system displays existing apartment projects and units.
	3	Staff selects Add New Apartment or Edit Apartment.
	4	Staff enters apartment details such as project, unit number, location and description.
	5	Staff enters price, size, floor, bedrooms, bathrooms, parking and facilities.
	6	Staff uploads apartment images, floor plans and other related media.
	7	Staff selects the apartment availability status.
	8	Staff submits the apartment details.
	9	The system validates the entered information.
	10	The system saves or updates the apartment listing.
	11	The system updates the current availability information.
	12	Customers and visitors can search, filter and view the updated apartment listing.
Extensions	**Step**	**Action**
	4a	If required apartment details are missing, the system asks the staff member to complete them.
	6a	If an uploaded file type is invalid, the system rejects the file and requests a valid file.
	8a	If staff cancels the operation, the system does not save the new changes.
	9a	If entered data is invalid, the system displays validation errors.
	11a	If the apartment becomes Reserved or Sold, the system updates its availability accordingly.
Open Issues:	Should customers be allowed to compare more than a fixed number of apartments at the same time?	

- IT25103871 Jathusayinie K.

Number	UC-05	
Name	Reservation Management	
Summary	Customers submit apartment reservation requests while staff verifies availability and approves or rejects reservations.	
Priority	5	
Preconditions	Customer is logged in, has completed an approved viewing appointment, and the selected apartment is Available.	
Postconditions	Reservation is recorded and the apartment availability status is updated.	
Primary Actor	Customer	
Secondary Actor	Staff, System	
Trigger	Customer selects Reserve Apartment for an available apartment.	
Main Scenario	**Step**	**Action**
	1	Customer opens the selected apartment details.
	2	Customer selects Reserve Apartment.
	3	The system verifies that the apartment is currently Available.
	4	The system displays reservation details and the reservation fee.
	5	Customer submits the reservation request.
	6	The system records the reservation with Pending status.
	7	Staff views the pending reservation request.
	8	Staff verifies the apartment availability and reservation details.
	9	Staff approves the reservation.
	10	The system changes the reservation status to Confirmed.
	11	The system temporarily holds the apartment for the customer.
	12	Customer uploads reservation payment evidence.
	13	Staff verifies the reservation payment.
	14	The system updates the reservation payment status.
	15	Customer can view the confirmed reservation and its current status.
Extensions	**Step**	**Action**
	3a	If the apartment is no longer available, the system prevents the reservation request.
	8a	If reservation details are invalid, staff rejects the reservation.
	9a	Staff may reject the reservation and the system changes its status accordingly.
	12a	If payment evidence is not submitted within two days, the reservation automatically expires.
	12a	When the reservation expires, the system restores the apartment status to Available.
	15a	If the customer or staff cancels the reservation, the system changes its status to Cancelled.
Open Issues:	Should the two-day reservation payment period be fixed or configurable by an administrator?	

- IT25103887 Kahanda I. M.

Number	UC-06	
Name	Booking and Viewing Appointment Management	
Summary	Customers request apartment viewing appointments while staff approves, rejects, reschedules or cancels appointment requests.	
Priority	5	
Preconditions	Customer is logged in and has selected an apartment to view.	
Postconditions	Viewing appointment is recorded with its current appointment status.	
Primary Actor	Customer	
Secondary Actor	Staff, System	
Trigger	Customer selects Book Viewing Appointment for an apartment.	
Main Scenario	**Step**	**Action**
	1	Customer opens the selected apartment details page.
	2	Customer selects Book Viewing Appointment.
	3	The system displays available dates and time slots.
	4	Customer selects a preferred date and time.
	5	Customer submits the appointment request.
	6	The system records the appointment with Pending status.
	7	Staff views the pending appointment request.
	8	Staff checks the requested date and time.
	9	Staff approves the appointment request.
	10	The system changes the appointment status to Confirmed.
	11	The system sends a confirmation notification to the customer.
	12	After the viewing, staff updates the appointment status to Completed.
Extensions	**Step**	**Action**
	3a	If no time slots are available, the system asks the customer to choose another date.
	8a	If the requested time is unavailable, staff reschedules the appointment.
	9a	Staff may reject the appointment request and provide a reason.
	10a	Customer or staff may cancel the appointment and the system sets the status to Cancelled.
	12a	If the customer does not attend the appointment, staff marks it as No-show.
Open Issues:	Should the system limit the number of viewing appointments a customer can book for the same apartment?	